

PROJECT TITLE:

Gene Expression Clustering and Pattern Discovery

OBJECTIVE:

To analyze Alzheimer's gene expression data using clustering techniques and identify co-expression patterns.

METHODOLOGY:

- Data preprocessing
- Data normalization
- PCA dimensionality reduction
- K-Means clustering
- Hierarchical clustering
- Heatmap visualization

RESULTS:

Genes were grouped into clusters based on similar expression patterns. Heatmaps and dendrograms revealed co-expression relationships among genes.

CONCLUSION:

The project successfully identified gene expression patterns associated with Alzheimer's disease using bioinformatics and machine learning approaches.